

MLE for the Binomial, Exponential and Normal Distributions

Contents

1. Overview	2
2. The binomial distribution	3
2.1 From probability to likelihood	3
2.2 The derivation with numbers	3
2.3 The general formula	4
3. The exponential distribution	4
3.1 The distribution of waiting times	4
3.2 The likelihood of all the waits	5
3.3 The derivation	6
4. The normal distribution	6
4.1 One mouse, then two	6
4.2 The log-likelihood	7
4.3 The MLE of the mean	8
4.4 The MLE of the standard deviation	8
5. Dividing by n or by $n - 1$	9
5.1 Two formulas for one spread	9
5.2 Biased but consistent	9
6. Summary	11
7. Sources	11
8. Key terms	11

1. Overview

Key point: Following the maximum likelihood recipe by hand gives three famous answers: the binomial's $\hat{p}$ is the share of successes, the exponential's $\hat{\lambda}$ is one over the average waiting time, and the normal's $\hat{\mu}$ and $\hat{\sigma}$ are the mean and standard deviation of the data.

This Note follows three explanations by Starmer (StatQuest, statquest.org): “Maximum Likelihood for the Binomial Distribution”, “Maximum Likelihood for the Exponential Distribution” and “Maximum Likelihood For the Normal Distribution”.

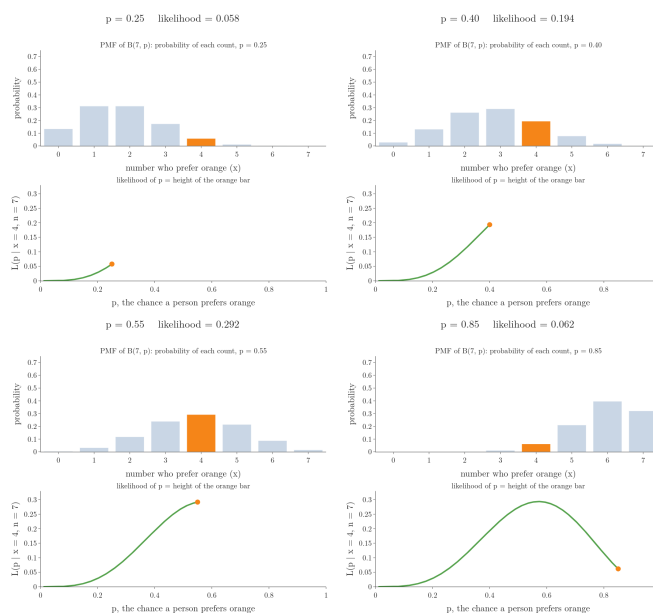


Figure 1: The PMF of $B(7, p)$ as p slides from 0.05 to 0.95. The orange bar is the probability of the observed count, 4 of 7; traced against p (bottom), it is the likelihood, highest at $p = 4/7$

The [maximum likelihood estimation Note](#) built the recipe: write the likelihood, take the log, differentiate, set the derivative to 0, solve. This Note runs it on three distributions, each in three steps (words, formula, small numbers):

- binomial: a share of yes answers (Section 2, Figure 1);
- exponential: waiting times (Section 3);
- normal: measurements such as weights (Section 4);
- why the normal's MLE variance divides by n , and how it differs from the sample variance with $n - 1$ (Section 5).

The Notebook checks every result by a grid search and against SciPy's built-in fitting.

2. The binomial distribution

Key point: If x of n trials are successes, the maximum likelihood estimate of the success probability is $\hat{p} = x/n$.

2.1 From probability to likelihood

Key point: The same binomial formula gives a probability when p is fixed and x varies, and a likelihood when x is fixed and p varies.

We ask 7 people whether they prefer orange or grape Fanta, and 4 say orange. Each answer is a Bernoulli trial, and the count of orange answers follows a binomial distribution $B(n, p)$ (see the [Bernoulli and binomial Note](#)):

$$P(x | n, p) = \binom{n}{x} p^x (1 - p)^{n-x}$$

With $p = 0.5$ fixed, this is the probability that 4 of 7 people prefer orange: $35 \times 0.5^4 \times 0.5^3 = 0.273$. Now fix the data, $n = 7$ and $x = 4$, and let p vary. The right-hand side stays the same formula; only the reading changes (the [probability vs likelihood Note](#)). We write it as $L(p | n = 7, x = 4)$.

p	0.25	0.5	0.57	0.75
$L(p n = 7, x = 4)$	0.058	0.273	0.294	0.173

Figure 1 sweeps p from left to right. For each p the top panel shows the whole PMF; the bar at $x = 4$ is the likelihood. Traced against p , the bar heights form the curve at the bottom, which peaks a little above 0.57.

2.2 The derivation with numbers

Key point: Log, differentiate, set to 0: $4/p - 3/(1 - p) = 0$ gives $p = 4/7$.

1. **In words:** take the log of the likelihood, differentiate with respect to p , set the result to 0 and solve for p .
2. **Formula:** the log turns the product into a sum and brings the exponents down:

$$\ell(p) = \log \binom{7}{4} + 4 \log p + 3 \log(1 - p)$$

The first term has no p in it, so its derivative is 0. The derivative of $\log p$ is $1/p$. For $\log(1 - p)$ the chain rule (see the [derivatives Note](#)) gives $1/(1 - p) \times (-1)$:

$$\frac{d\ell}{dp} = \frac{4}{p} - \frac{3}{1 - p}$$

3. **Example:** set it to 0 and multiply both sides by $p(1 - p)$:

$$4(1 - p) - 3p = 0 \implies 4 - 7p = 0 \implies \hat{p} = \frac{4}{7} = 0.571$$

The likelihood there is 0.294, higher than at every p in the table.

2.3 The general formula

Key point: The same steps with letters give $\hat{p} = x/n$: successes divided by trials.

1. **In words:** for x successes in n trials, the MLE of p is the observed share of successes.
2. **Formula:** the log-likelihood and its derivative are

$$\ell(p) = \log \binom{n}{x} + x \log p + (n - x) \log(1 - p), \quad \frac{d\ell}{dp} = \frac{x}{p} - \frac{n - x}{1 - p}$$

Setting the derivative to 0 and multiplying by $p(1 - p)$: $x(1 - p) - (n - x)p = 0$. The terms $-xp$ and $+xp$ cancel, leaving $x - np = 0$:

$$\hat{p} = \frac{x}{n}$$

3. **Example:** 4 orange answers out of 7 give $4/7 = 0.571$; 40 out of 70 give the same 0.571.

The second derivative, $-x/p^2 - (n - x)/(1 - p)^2$, is negative for every p between 0 and 1, so this is a maximum.

The answer looks obvious once known: the best guess of a probability is the observed proportion. The derivation turns the intuition into a proof.

Extra: With 40 of 70 the peak sits at the same 0.571, but the likelihood curve is much narrower: more data pins p down more tightly. The Notebook draws both curves.

3. The exponential distribution

Key point: For waiting times $x_1, \dots, x_n$, the MLE of the rate is $\hat{\lambda} = n / \sum x_i = 1/\bar{x}$: one over the average wait.

3.1 The distribution of waiting times

Key point: $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. The rate λ is how many events happen per unit time on average; the average wait is $1/\lambda$.

The **exponential distribution** models the time between random events: the wait for the next text message, the time until the next person opens a web page. The distribution is right-skewed: short waits are common, long waits rare. The [sampling distribution and CLT Note](#) used it as a skewed population.

1. **In words:** the density starts at height λ at $x = 0$ and falls off exponentially. The **rate parameter** λ is the average number of events per unit of time.
2. **Formula:**

$$f(x | \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0$$

3. **Example:** with $\lambda = 2$ events per second, on average one event every $1/2$ second; the density at $x = 1$ second is $2e^{-2} = 0.27$. With $\lambda = 0.5$, one event every 2 seconds, and the density at 1 second is $0.5e^{-0.5} = 0.30$.

Figure 2 (left) shows the three curves for $\lambda = 0.5, 1$ and 2 . A large rate means a tall start and a fast drop: short waits.

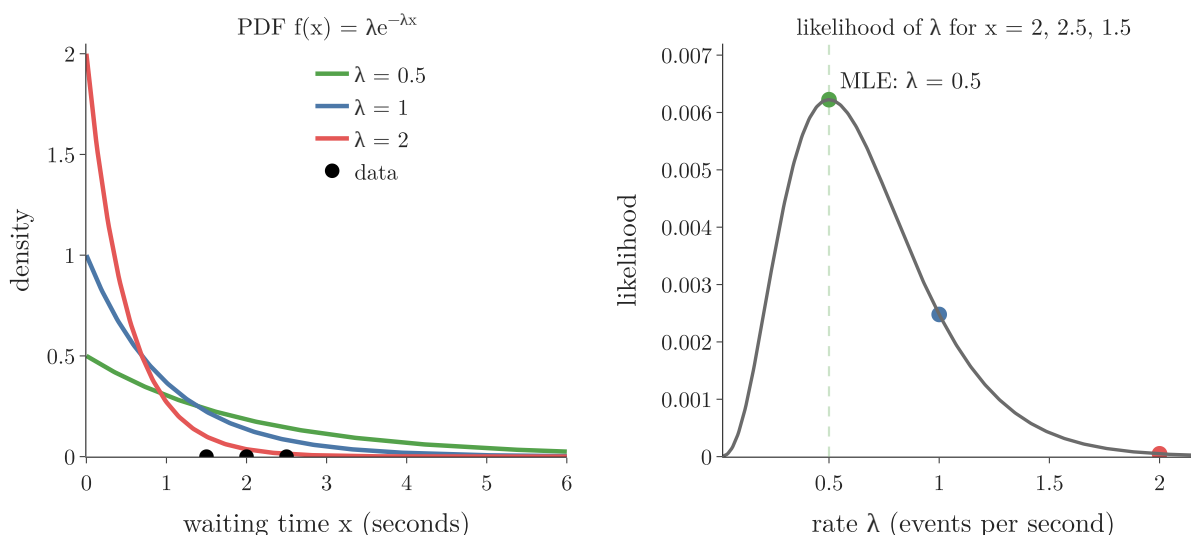


Figure 2: Left: exponential PDFs for three rates and the three waiting times. Right: the likelihood of the rate for those waiting times, highest at 0.5

3.2 The likelihood of all the waits

Key point: Multiplying the n densities collects the λ 's into λ^n and the exponents into $-\lambda \sum x_i$.

1. **In words:** each waiting time contributes the height of the curve above it. For independent waits the heights multiply, as in the [maximum likelihood estimation Note](#).
2. **Formula:** pull the n factors λ out of the product and add the exponents ($e^a e^b = e^{a+b}$):

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda(x_1 + \dots + x_n)}$$

3. **Example:** three waits between views of a web page: $x_1 = 2, x_2 = 2.5, x_3 = 1.5$ seconds, with sum 6. For $\lambda = 1$, $L = 1^3 e^{-6} = 0.0025$; for $\lambda = 0.5$, $L = 0.5^3 e^{-3} = 0.0062$.

3.3 The derivation

Key point: $\ell(\lambda) = n \log \lambda - \lambda \sum x_i$; its derivative $n/\lambda - \sum x_i$ is 0 at $\hat{\lambda} = n / \sum x_i$.

1. **In words:** the log turns λ^n into $n \log \lambda$ and the exponential into its exponent. Then differentiate and set to 0.

2. **Formula:**

$$\ell(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n x_i, \quad \frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0 \implies \hat{\lambda} = \frac{n}{\sum_i x_i} = \frac{1}{\bar{x}}$$

3. **Example:** $\hat{\lambda} = 3/(2 + 2.5 + 1.5) = 3/6 = 0.5$ events per second. The average wait is 2 seconds, so the rate is one event every 2 seconds.

The second derivative is $-n/\lambda^2 < 0$, so this is the peak of Figure 2 (right). The fitted model is the green curve on the left, $f(x) = 0.5e^{-0.5x}$.

Python: SciPy's `fit` method uses maximum likelihood by default (its documentation says so).

```
import numpy as np
from scipy import stats
```

```
waits = np.array([2, 2.5, 1.5])
# floc=0 fixes the start of the curve at 0
loc, scale = stats.expon.fit(waits, floc=0)
scale          # 2.0: SciPy uses scale = 1 / lambda
1 / scale      # 0.5, the same as 3 / 6
```

SciPy describes the exponential by its **scale**, the average wait $1/\lambda$, rather than by the rate (`scipy.stats.expon` documentation). SciPy also has a shift parameter `loc`; without `floc=0` SciPy fits the shift too, and the Notebook shows the curve's start then moves to the smallest wait, 1.5.

4. The normal distribution

Key point: The MLE of μ is the mean of the data, and the MLE of σ is the standard deviation computed with n in the denominator.

4.1 One mouse, then two

Key point: With one observation (one recorded value) the likelihood is the height of the curve above it; with two, the product of two heights.

The normal PDF (see the [normal distribution Note](#)) has two parameters, the mean μ (location) and the standard deviation σ (width):

$$f(x \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

We weigh one mouse: 32 grams. With $\sigma = 2$ fixed, the likelihood of $\mu = 28$ is $f(32 | 28, 2) = 0.027$; of $\mu = 30$ it is 0.121; of $\mu = 32$ it is 0.199, the highest. With one point, the best curve is centred on it.

With two mice, 32 and 34 grams, the likelihood of a curve is the product of its two heights, because the weighings are independent. For $\mu = 28$, $\sigma = 2$: $0.027 \times 0.0022 = 6.0 \times 10^{-5}$. With n mice it is a product of n heights.

Extra: One point is not enough to estimate σ . With μ on the single point, the likelihood is the height of the curve at its own centre, $f(x | x, \sigma) = 1/(\sigma\sqrt{2\pi})$, which grows without limit as $\sigma \rightarrow 0$: there is no best σ . The Notebook prints 0.20, 0.80, 3.99 and 39.89 for $\sigma = 2, 0.5, 0.1$ and 0.01. The same runaway returns in the [Gaussian mixture models Note](#).

4.2 The log-likelihood

Key point: The log of one normal density is $-\frac{1}{2}\log(2\pi) - \log\sigma - (x - \mu)^2/(2\sigma^2)$; summing over n points gives the log-likelihood.

1. **In words:** the log splits each density into three pieces: a constant, minus $\log\sigma$, and minus the squared distance from the mean divided by $2\sigma^2$.
2. **Formula:** using $\log(1/a) = -\log a$, $\log\sqrt{a} = \frac{1}{2}\log a$ and $\log e^b = b$,

$$\log f(x_i | \mu, \sigma) = -\frac{1}{2}\log(2\pi) - \log\sigma - \frac{(x_i - \mu)^2}{2\sigma^2}$$

Adding the n terms, the first two pieces appear n times:

$$\ell(\mu, \sigma) = -\frac{n}{2}\log(2\pi) - n\log\sigma - \frac{1}{2\sigma^2}\sum_{i=1}^n(x_i - \mu)^2$$

3. **Example:** for the five mice of the [maximum likelihood estimation Note](#), 29, 31, 32, 33, 35, at $\mu = 32$, $\sigma = 2$: the squared distances are 9, 1, 0, 1, 9, with sum 20, so

$$\ell = -2.5 \times 1.838 - 5 \times 0.693 - \frac{20}{8} = -4.595 - 3.466 - 2.5 = -10.56$$

Figure 3 shows ℓ over both parameters as a contour map. The map has a single peak. The dashed lines are the two one-parameter searches of the [maximum likelihood estimation Note](#); both cross at the peak.

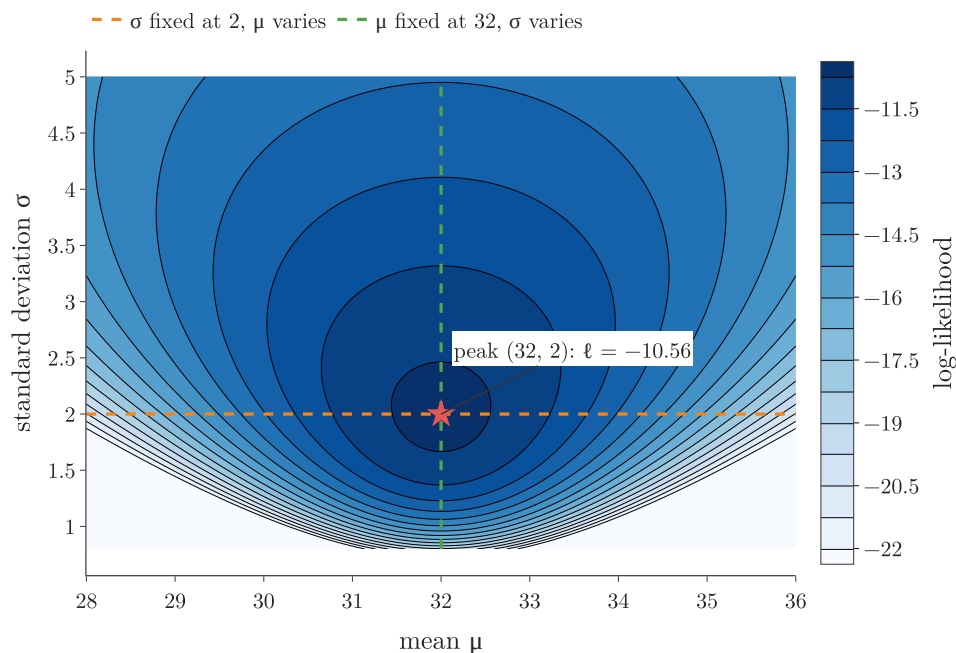


Figure 3: Contour map of the normal log-likelihood of the five mouse weights over the mean and the standard deviation, with a single peak at (32, 2)

4.3 The MLE of the mean

Key point: Differentiating with respect to μ (with σ held constant) and setting to 0 gives $\hat{\mu} = \bar{x}$.

1. **In words:** only the last term of ℓ contains μ . Treat σ as a constant, take the partial derivative with respect to μ (see the [partial derivatives and gradients Note](#)), set it to 0.
2. **Formula:** by the chain rule, the derivative of $(x_i - \mu)^2$ with respect to μ is $2(x_i - \mu) \times (-1)$, so

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = \frac{1}{\sigma^2} \left(\sum_i x_i - n\mu \right) = 0 \quad \implies \quad \hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

3. **Example:** $\hat{\mu} = (29 + 31 + 32 + 33 + 35)/5 = 160/5 = 32$ grams.

The answer does not depend on σ : whatever the width, the best centre is the average.

4.4 The MLE of the standard deviation

Key point: Differentiating with respect to σ (with μ held at $\hat{\mu}$) gives $\hat{\sigma}^2 = \sum (x_i - \bar{x})^2 / n$.

1. **In words:** now treat μ as a constant. Two terms of ℓ contain σ : $-n \log \sigma$ and the sum divided by $2\sigma^2$.

2. **Formula:** the derivative of $-n \log \sigma$ is $-n/\sigma$. Writing $1/(2\sigma^2)$ as $\frac{1}{2}\sigma^{-2}$, its derivative is $-\sigma^{-3}$, so the minus sign in front makes the second term positive:

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

Multiply by σ^3 , move $n\sigma^2$ to the other side, divide by n , and put in $\mu = \hat{\mu}$:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad \hat{\sigma} = \sqrt{\hat{\sigma}^2}$$

3. **Example:** the squared distances from 32 add up to 20, so $\hat{\sigma}^2 = 20/5 = 4$ and $\hat{\sigma} = 2$ grams.

So the maximum likelihood normal curve for the mice is $N(32, 2^2)$, exactly where the searches in the [maximum likelihood estimation Note](#) peaked.

```
Python: stats.norm.fit returns the two maximum likelihood estimates.
mice = np.array([29, 31, 32, 33, 35])
stats.norm.fit(mice)      # (32.0, 2.0)
mice.mean(), mice.std()  # (32.0, 2.0): NumPy's std divides by n
mice.std(ddof=1)         # 2.236: divides by n - 1
```

5. Dividing by n or by $n - 1$

Key point: The MLE variance divides by n and is slightly too small on average; the sample variance divides by $n - 1$ and is right on average. The gap vanishes as n grows.

5.1 Two formulas for one spread

Key point: For the five mice, the MLE variance is 4 and the sample variance is 5.

The [measures of dispersion Note](#) (Section 4.3) defined the sample variance s^2 with $n - 1$ in the denominator, Bessel's correction. Maximum likelihood gives n instead.

1. **In words:** both formulas add the squared distances from the sample mean; they divide by different counts.
2. **Formula:**

$$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{n}{n-1} \hat{\sigma}_{\text{ML}}^2$$

3. **Example:** the sum is 20. $\hat{\sigma}_{\text{ML}}^2 = 20/5 = 4$; $s^2 = 20/4 = 5$.

5.2 Biased but consistent

Key point: On average the MLE variance equals $(n-1)/n \times \sigma^2$, so it is biased low; since $(n-1)/n \rightarrow 1$, it is still consistent.

An estimator is **unbiased** if its average over many samples equals the true value, and **biased** if it is systematically off. The intuition: the sample mean $\bar{x}$ sits in the middle of its own sample, so the observations are always a little closer to $\bar{x}$ than to the true mean μ , and squared distances measured from $\bar{x}$ come out a little too small. The identity $\sum_i (x_i - \bar{x})^2 = \sum_i (x_i - \mu)^2 - n(\bar{x} - \mu)^2$ makes the gap exact. The measures of dispersion Note showed by simulation that dividing by n comes out too small on average. The size of the gap is a simple factor:

1. **In words:** averaged over many samples of size n , the MLE variance is the true variance times $(n-1)/n$.

2. **Formula:**

$$\text{average of } \hat{\sigma}_{\text{ML}}^2 = \frac{n-1}{n} \sigma^2, \quad \text{average of } s^2 = \sigma^2$$

3. **Example:** for samples of 5 mice from a population with $\sigma^2 = 4$, the MLE variance averages $4/5 \times 4 = 3.2$, while s^2 averages 4. With $n = 100$ the factor is 0.99, and the difference hardly matters. The Notebook draws 100,000 samples of 5 and gets 3.21 and 4.01.

Extra: Why the factor is $(n-1)/n$. Write $x_i - \mu = (x_i - \bar{x}) + (\bar{x} - \mu)$, square, and add over i . The cross term $2(\bar{x} - \mu) \sum_i (x_i - \bar{x})$ is 0, because the distances from the mean add up to 0. So

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i - \mu)^2 - n(\bar{x} - \mu)^2$$

Averaged over many samples, each $(x_i - \mu)^2$ is σ^2 , and $(\bar{x} - \mu)^2$ is the variance of the sample mean, σ^2/n (the [sampling distribution and CLT Note](#)). The right-hand side averages $n\sigma^2 - \sigma^2 = (n-1)\sigma^2$; dividing by n gives the factor.

So maximum likelihood does not promise an unbiased estimate. Maximum likelihood promises the parameters that make the observed data most likely, and, as n grows, an estimate that converges to the truth (consistency, from the [maximum likelihood estimation Note](#)).

6. Summary

Distribution	Parameter	Log-likelihood ℓ	MLE	Example
Binomial $B(n, p)$	p	$x \log p + (n - x) \log(1 - p) + \text{const}$	$\hat{p} = x/n$	4 of 7: 0.571
Exponential	λ	$n \log \lambda - \lambda \sum x_i$	$\hat{\lambda} = 1/\bar{x}$	waits 2, 2.5, 1.5: 0.5
Normal	μ	$-n \log \sigma - \sum (x_i - \mu)^2 / (2\sigma^2) + \text{const}$	$\hat{\mu} = \bar{x}$	mice: 32
Normal	σ	same	$\hat{\sigma}^2 = \sum (x_i - \bar{x})^2 / n$	mice: $\hat{\sigma} = 2$

- Every derivation follows the same recipe: product of densities, log, derivative, set to 0, solve.
- The binomial MLE is the observed share of successes.
- The exponential distribution $\lambda e^{-\lambda x}$ models waiting times; its MLE rate is one over the average wait.
- The normal MLEs are the sample mean and the standard deviation with n in the denominator.
- The MLE variance is biased low by the factor $(n - 1)/n$; the sample variance with $n - 1$ is unbiased. Both agree for large n .
- SciPy's `fit` methods compute maximum likelihood estimates by default.

7. Sources

- Starmer, J. “Maximum Likelihood for the Binomial Distribution”, StatQuest (statquest.org). Used in Section 2.
- Starmer, J. “Maximum Likelihood for the Exponential Distribution”, StatQuest (statquest.org). Used in Section 3.
- Starmer, J. “Maximum Likelihood For the Normal Distribution”, StatQuest (statquest.org). Used in Section 4.
- SciPy documentation: `scipy.stats.expon` (scale = $1/\lambda$) and `rv_continuous.fit` (maximum likelihood by default). Used in Section 3.3.

8. Key terms

Term	Meaning
Exponential distribution	The distribution of waiting times between random events, $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$

Term	Meaning
Rate parameter λ	The average number of events per unit time; the average wait is $1/\lambda$
Scale (in SciPy)	SciPy's parameter for the exponential distribution, the average wait $1/\lambda$
MLE of a binomial p	$\hat{p} = x/n$, the observed share of successes
MLE of an exponential rate	$\hat{\lambda} = n / \sum x_i = 1/\bar{x}$
MLE of a normal distribution	$\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \sum (x_i - \bar{x})^2 / n$
Unbiased estimator	An estimator whose average over many samples equals the true value
Biased estimator	An estimator that is systematically too high or too low on average